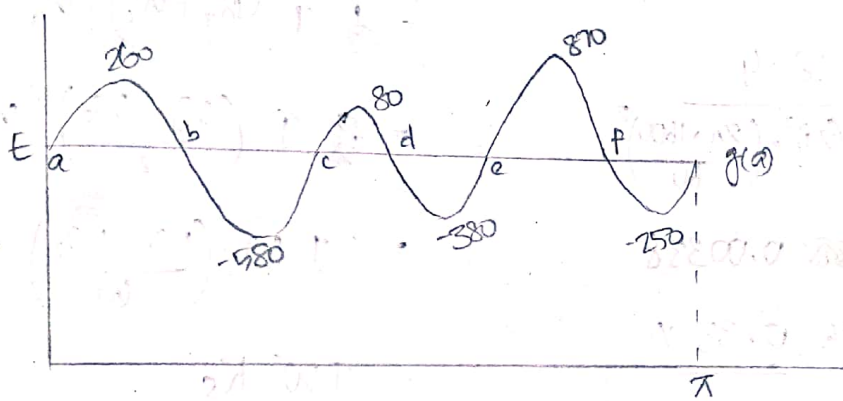


Q) The turning moment diagram for a petrol engine is drawn to a vertical scale of $1 \text{ mm} = 500 \text{ Nm}$ and a horizontal scale of $1 \text{ mm} = 3^\circ$. The turning moment dia repeats itself after every half rev of crankshaft. The areas above and below the mean torque line are $260 - 580 - 80 - 380 - 870 - 250 \text{ mm}^2$. Rotating parts have a mass of 55 kg and the radius of gyration is 2.1 m . If the engine speed is 1600 rpm . Determine the coeff. of fluctuation of speed.



Energy at $a = E$

$b = E + 260$

$c = E + 260 - 580$

$= E - 320$

$d = E - 320 + 80$

$= E - 240$

$e = E - 240 - 380$

$= E - 620$

$f = E - 620 + 870$

$= E - 250$

$$1 \text{ mm}^2 = 500 \text{ Nm} \times 3 \times \frac{\pi}{180} = f$$

$$g = E + 250 - 250$$

$$= E$$

$$L_{man} = E + 260 - (E - 620)$$

$$= 880 \text{ units}$$

$$= 880 \times 500 \times \frac{3}{180} \pi \text{ Nm} = \underline{\underline{23038.4 \text{ Nm}}}$$

$$K_s = \frac{L_{man}}{I \omega^2}$$

$$\omega = \frac{\omega_1 + \omega_2}{2}$$

$$= \frac{23038.4}{55 \times (2.1)^2 \times \left(\frac{2\pi \times 1600}{60}\right)^2}$$

$$= 0.0338$$

$$= \underline{\underline{0.33\%}}$$

$$L_{man} = \frac{1}{2} I \omega_1^2 - \frac{1}{2} I \omega_2^2$$

$$= \frac{1}{2} I (\omega_1^2 - \omega_2^2)$$

$$= \frac{1}{2} I (\omega_1 + \omega_2) (\omega_1 - \omega_2)$$

$$= I \left(\frac{\omega_1 + \omega_2}{2} \right) (\omega_1 - \omega_2)$$

$$= I \omega \left(\frac{\omega_1 - \omega_2}{\omega} \right) \omega$$

$$= I \omega^2 K_s$$

$$K_s = \left(\frac{\omega_1 - \omega_2}{\omega} \right)$$

For the above braking moment dia what should be the ms of flywheel if the speed should be maintained within the limit $1600 \pm 50 \text{ rpm}$.

$$K_s = \frac{100}{1600}$$

$$I = \frac{L_{man}}{K_s \times \omega^2} = \frac{23038.4}{0.0625 \times \left(\frac{2\pi \times 1600}{60}\right)^2} = \underline{\underline{13.13 \text{ kgm}^2}}$$

$$\omega_1 = 1650 \text{ rpm}$$

$$\omega_2 = 1550 \text{ rpm}$$

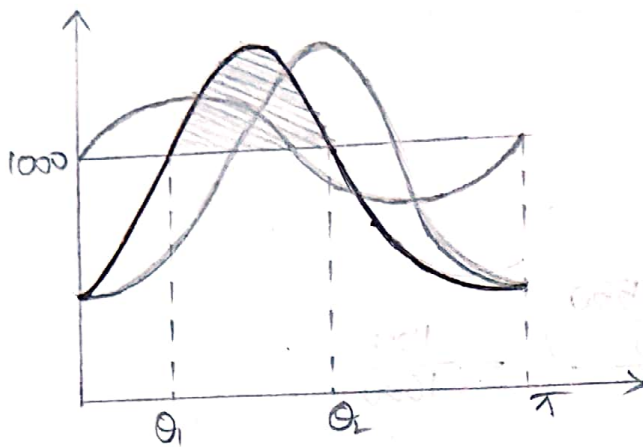
$$k = \frac{\omega_1 - \omega_2}{\omega}$$

$$= \frac{\left(\frac{2\pi \times 1650}{60} - \frac{2\pi \times 1550}{60} \right)}{\left(\frac{2\pi \times 1600}{60} \right)} = \frac{100}{1600}$$

2) The torque delivered by a 2 stroke engine is represented by $T = 1000 + 300 \sin 2\theta - 500 \cos 2\theta$ Nm, where θ is the angle turned by the crank from the inner dead centre. The engine speed is 250 rpm. Mass of the flywheel is 400 kg and radius of gyration is 100 mm. Determine

- (i) Power developed
- (ii) Total % fluctuation of speed
- (iii) Ang. accelⁿ of flywheel when the crank has rotated to an angle of 60° from the IDC
- (iv) Max ang accelⁿ and retardation of the flywheel.

$$T = (1000 + 300 \sin 2\theta - 500 \cos 2\theta) \text{ Nm}$$



$$i) P = T_{\text{mean}} \times \omega = 1000 \times \frac{2\pi \times 290}{60}$$

$$= 26179.9 \approx \underline{26180 \text{ W}}$$

$$T_{\text{mean}} = \frac{\int_0^\pi T d\theta}{\pi}$$

$$\Delta T = T - T_{\text{mean}}$$

$$\int_{\theta_1}^{\theta_2} \Delta T d\theta = E_{\text{man}}$$

$$K_s = \frac{E_{\text{man}}}{I \omega^2}$$

$$T = T_{\text{mean}}$$

$$1000 + 300 \sin 2\theta - 500 \cos 2\theta = 1000$$

$$300 \sin 2\theta = 500 \cos 2\theta$$

$$\tan 2\theta = \frac{5}{3}$$

$$2\theta = \underline{59.03^\circ}, \underline{239.03^\circ}$$

$$\theta = 29.51^\circ, 119.51^\circ$$

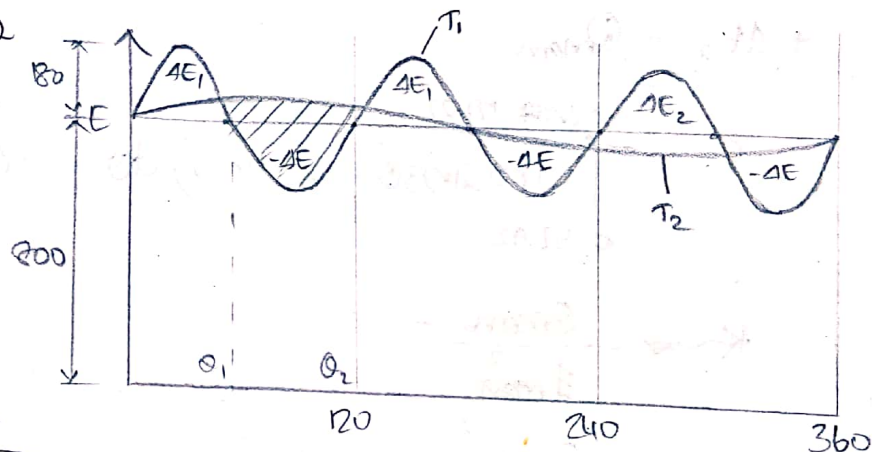
$$E_{\text{man}} = \int_{29.51}^{119.51} \Delta T d\theta = E_{\text{man}}$$

$$\begin{aligned}
 P_{\text{man}} &= \int_{29.51}^{119.51} (300 \sin 2\theta - 500 \cos 2\theta) d\theta \\
 &= \left(-150 \cos 2\theta - 250 \sin 2\theta \right) \Big|_{29.51}^{119.51} \\
 &= \underline{583.09}
 \end{aligned}$$

$$\begin{aligned}
 K_s &= \frac{P_{\text{man}}}{I \omega^2} = \frac{583.1}{400 \times (0.4)^2 \times \left(\frac{2\pi \times 2500}{60} \right)^2} \\
 &= 0.01329 = \underline{\underline{1.329}}
 \end{aligned}$$

2) A machine is coupled to a 2 stroke engine which produces a torque of $800 + 180 \sin 3\theta$ Nm. where θ is the crank angle. The mean engine speed is 400 rpm. The flywheel and the other rotating parts attached to the engine have a mass of 350 kg at a radius of gyration 220 mm. Calculate the power of the engine and the total fluctuation of speed of the flywheel. ~~There is a still torque is~~
 The resisting torque is $800 + 80 \sin \theta$ Nm.

4) $T_1 = 800 + 180 \sin 3\theta$
 $T_2 = 800 + 80 \sin \theta$



$$\frac{180}{80} = \frac{\sin \theta}{\sin 3\theta} = 2.25$$

$$\frac{\sin \theta}{4 \sin 3\theta - 3 \sin \theta} = 2.25$$

$$\frac{\sin \theta}{3 \sin \theta - 4 \sin 3\theta} = 2.25$$

$$\sin \theta = 2.25 (3 \sin \theta - 4 \sin 3\theta)$$

$$800 + 180 \sin 3\theta = 800 + 80 \sin \theta \leftarrow (\tau_1 = \tau_2)$$

$$180 (3 \sin \theta - 4 \sin^3 \theta) = 80 \sin \theta$$

$$180 \sin \theta (3 - 4 \sin^2 \theta) = 80 \sin \theta$$

$$\sin^2 \theta = 0.637$$

$$\sin \theta = \pm 0.799$$

$$0, 52.92^\circ, 127.02^\circ, 180^\circ, 232.92^\circ, 307.02^\circ$$

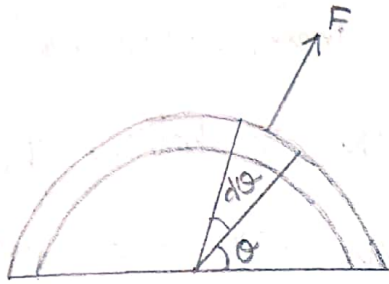
$$e = \int A T d\theta$$

$$+4E_2 = Q_{\text{man}}$$

$$e = \int_{52.92}^{307.02} (180 \sin 3\theta - 80 \sin \theta) d\theta = \underline{\underline{208.3 \text{ Nm}}}$$

$$K = \frac{Q_{\text{man}}}{I_w^2}$$

$$K = \frac{Q_{\text{man}}}{I_w^2} = \frac{208.3}{350 \times 0.2^2 \times \left(\frac{2\pi \times 400}{60} \right)^2} = 0.007 = \underline{\underline{0.7\%}}$$



$$\int_0^\pi P(R d\theta) R \omega^2 \sin \theta$$

$$F = m R \omega^2$$

$$= (P R d\theta A) R \omega^2$$

$$\text{vertical force} = \int_0^\pi (dF \sin \theta)$$

$$= \int_0^\pi P R d\theta A R \omega^2 \sin \theta$$

$$= P R^2 \omega^2 A \int_0^\pi \sin \theta d\theta$$

$$= [-\cos \theta]_0^\pi P R^2 \omega^2 A$$

$$= (1 - (-1)) P R^2 \omega^2 A$$

$$= 2 P R^2 \omega^2 A$$

$$2 P R^2 \omega^2 A = \sigma A L$$

$$\sigma = P R^2 \omega^2$$

$$= \rho v^2$$

2) The turning moment dia of multicylinder engine is shown by a vertical scale of $1 \text{ mm} = 650 \text{ Nm}$. Horizontal scale of $1 \text{ mm} = 4.5^\circ$. Areas above & below the mean torque line are $-2, +380, -260, +310, -300, +242, -380, +265$ and -229 mm^2 . The fluctuation of speed is limited to $\pm 1.8\%$ of mean speed which is 400 rpm . The density of the material of the ring is 7000 kg/m^3 and width of ring is 4.5 times its thickness. The centrifugal stress in ring material is limited to 6 N/mm^2 . If the boss & the arms contribute 5% of M.O.I. Determine the dia & C.S. of Flywheel

$$E_{\text{mean}} = 22360 \text{ Nm}$$

$$\text{Fluctuation} = \pm 0.018$$

$$K_s = 0.036$$

$$\sigma = 6 \text{ N/mm}^2 = 6 \times 10^6 \text{ N/m}^2$$

$$\rho = 7000 \text{ kg/m}^3 \times V^2 (\text{m/s})^2$$

$$V = 29 \text{ m/s}$$

$$V = r\omega$$

$$r = V/\omega = \frac{29}{\left(\frac{2\pi \times 400}{60}\right)} = 0.7 \text{ m}$$

$$I = \frac{E_{\text{mean}}}{K_s \times \omega^2}$$

$$= \frac{22360}{0.036 \times \left(\frac{2\pi \times 400}{60}\right)^2} = 354 \text{ kg/m}^2$$

~~mean fluctuation~~

$$\text{Mean} = 22360 \text{ Nm}$$

$$\text{Fluctuation} = \pm 0.018$$

$$K_s = 0.036$$

$$I_{\text{rim}} = 0.95 \times 354$$

$$= 336.3 \text{ kgm}^2$$

$$R = 0.7 \text{ m}$$

$$I_{\text{rim}} = M_{\text{rim}} R^2$$

$$M_{\text{rim}} = \frac{I_{\text{rim}}}{R^2} = \frac{336.3}{0.49}$$

$$= 686.32 \text{ kg}$$

$$M_{\text{rim}} = 2\pi R A \rho$$

$$A = \frac{M_{\text{rim}}}{2\pi R \rho} = \frac{686.32}{2\pi \times 0.7 \times 700} = \frac{686.32}{2\pi \times 7 \times 700} = 0.22$$

$$= W = 4.5 \text{ t}^2$$

$$t = 0.01 \text{ m}$$

Gyroscope

$$\frac{d(I\omega)}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta(I\omega)}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} I\omega \times \frac{\Delta\omega}{\Delta t}$$

$$= I\omega \omega_p$$

$$I\omega\omega_p = C = mgl$$

